When Does Detect-and-Reweight Work for Synthetic Contamination?

Anonymous ACL submission

Abstract

As synthetic text pervades web corpora, detect-and-reweight methods—downweighting likely-synthetic samples during training—have emerged to prevent model collapse, yet when they actually help remains unclear. We investigate this question through controlled continued-pretraining experiments (Qwen3-0.6B, 132M tokens, 5 seeds) and identify three effectiveness conditions. First, published detect-and-remove evaluations are confounded by multi-epoch exposure: over 95% of the reported advantage vanishes under epoch-matched controls. Second, the detector must provide both sufficient discrimination *and* good calibration; high-AUC detectors with poor calibration degrade performance below uniform training. Third, minimum intervention—applying the smallest reweighting exponent that meets a quality floor—outperforms unconstrained weight maximization. As an illustrative application of the minimum-intervention principle, we derive a dual-constrained adaptive exponent b^* that improves over uniform training at contamination ratios $\rho \geq 0.4$, reaching conventional significance after multiple-comparison correction, while requiring no per-model tuning. These findings apply within our self-contamination setup at 0.6B scale; generalization to cross-model contamination and larger models remains to be established.

1 Introduction

The detect-and-reweight paradigm (Drayson et al., 2025) has emerged as a response to synthetic contamination in web corpora, now estimated at 20–60% of crawled data (Cheng et al., 2025). Training on such data risks model collapse (Shumailov et al., 2024; Dohmatob et al., 2025), motivating methods that downweight likely-synthetic samples via $w(x) = (1 - q(x))^b$, where $q(x)$ is a detector's synthetic probability and b is a sharpening exponent. An alternative, Detect and Remove (D&R), hard-filters samples above a threshold (Feng et al., 2025).

Despite growing adoption, the paradigm's effectiveness conditions are poorly characterized. Does D&R genuinely outperform soft reweighting, or do published comparisons confound filtering quality with multi-epoch exposure? What detector properties are necessary for effective reweighting? How does the reweighting mechanism behave across contamination levels, and can b be set without per-model grid search?

We investigate these questions through controlled experiments on Qwen3-0.6B continued pretraining (132M FineWeb tokens, ρ=0.4, 5 seeds), with extensions to $\rho \in \{0.2, 0.6\}$ and 1.7B parameters (§7 discusses scope). To ground the analysis of exponent selection, we implement a dual-constrained formulation as an illustrative application: find the minimum b such that $\alpha_{\text{eff}}(b) \geq \tau$ (quality floor, motivated by convergence theory (Wang et al., 2026)), subject to $\text{ESS}(b) \geq n_{\min}$ (stability ceiling) (Kish, 1965; Vehtari et al., 2024).

Our analysis yields three findings:

- **Epoch confound in D&R evaluation.** The bulk of D&R's perplexity (PPL) advantage over adaptive reweighting (>95%) is attributable to multi-epoch exposure, not filtering quality; epoch-matched controls (N=3) reduce the gap from 0.394 to Δ=0.017 (Table 2).
- **Detector requirements for effective reweighting.** Both sufficient discrimination (area under the ROC curve; AUC) and good calibration (expected calibration error; ECE) are empirically observed requirements in our self-contamination setting; a high-AUC but poorly calibrated detector degrades performance below uniform (3 detectors, 3 seeds; §5.2).
- **Minimum intervention outperforms maximization.** An α_{eff} quality constraint binds in ∼50% of batches at ≈5.9× lower sharpening exponents than an ESS-only alternative (§5.3); at

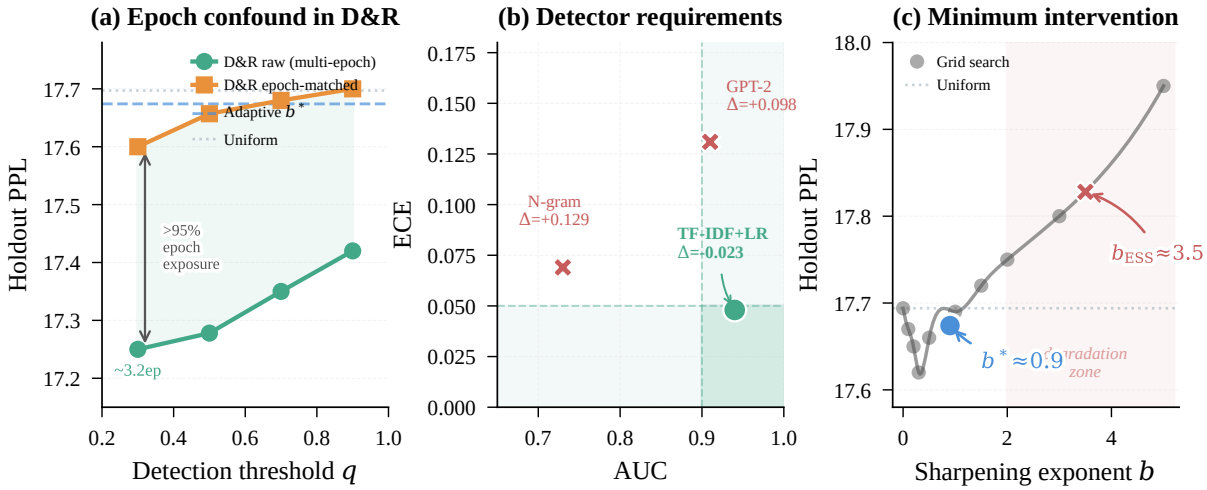


Figure 1: Three effectiveness conditions for detect-and-reweight. **(a)** D&R's PPL advantage largely vanishes under epoch-matched controls: the green–orange gap is multi-epoch exposure, not filtering quality (§5). **(b)** Only detectors with both high AUC *and* low ECE fall in the effective zone; comparable AUC with poor calibration degrades PPL below uniform. **(c)** Adaptive $b^*\approx 0.9$ operates near the grid-search optimum, while ESS-only $b\approx 3.5$ falls in the degradation zone.

ρ=0.6, ESS-only collapses toward uniform while the dual constraint maintains its advantage.

To demonstrate the minimum-intervention principle, we derive a dual-constrained adaptive b^* that eliminates per-model grid search ($\Delta=-0.023$ at ρ=0.4, -0.032 at ρ=0.6; Table 3), with transferable defaults (τ=0.7, $n_{\min}=\lfloor B/2 \rfloor$). The method fails at ρ=0.2; at 1.7B it shows no degradation but the benefit diminishes.

2 Related Work

Synthetic Contamination and Model Collapse. Training on recursively generated data leads to irreversible quality degradation (*model collapse*) (Shumailov et al., 2024). Alemohammad et al. (2024) formalize this as Model Autophagy Disorder, showing that quality inevitably degrades without fresh real data. More critically, Dohmatob et al. (2025) prove under scaling-law regimes that even a vanishingly small synthetic fraction suffices to trigger collapse, ruling out data scaling as a remedy. Cheng et al. (2025) confirm that synthetic content now constitutes an estimated 20–60% of web corpora. Not all synthetic data is harmful: the Phi series (Gunasekar et al., 2023; Li et al., 2023) demonstrates that curated synthetic data can improve training efficiency, but this assumes intentional generation with quality control, unlike the uncontrolled web contamination we address.

Detection, Filtering, and Data Reweighting. Multiple detectors produce continuous scores suitable for downstream weighting: DetectGPT (Mitchell et al., 2023) uses probability curvature, Binoculars (Hans et al., 2024) cross-perplexity ratios, and Min-k% Prob (Shi et al., 2024) low-probability token statistics; Wu et al. (2025) survey the broader detection landscape. A parallel line of work optimizes data composition at the domain (Xie et al., 2023a,b; Ye et al., 2025) or instance level (Abbas et al., 2023; Bengio et al., 2009; Mindermann et al., 2022; Sow et al., 2025), using loss, frequency, quality, or domain membership as the selection signal; none targets synthetic provenance.

Most directly related to our work, Drayson et al. (2025) propose $w_j = (1-q_j)^b$ for collapse prevention via detect-and-reweight in a from-scratch pretraining setting, where q_j is the detector's synthetic probability. We adopt the same weighting formula but differ in several fundamental ways. **(i)** Drayson et al. select b via validation-set grid search and fix it throughout training; different model scales require different values (b=10 for $\leq$360M, b=1 for 1.7B), so the choice is *per-model* and expensive. Our b^* is derived per-batch from dual constraints ($\alpha_{\text{eff}} \geq \tau$, ESS $\geq n_{\min}$) at zero validation cost. **(ii)** Their sampling importance resampling (SIR) discretizes the weight distribution; we apply direct loss reweighting with continuous gradients. **(iii)** They employ a ModernBERT-150M detector (AUC = 0.986) on WikiText-2 ($\sim$2M tokens); we show that a lightweight TF-IDF + logis-

tic regression (LR) detector (AUC = 0.94) suffices on FineWeb-scale data (132M+ tokens), and both calibration and sufficient discrimination appear required in our setting for effective reweighting.

Theoretical Foundations. He et al. (2025) derive the optimal real-data weight under known mixture proportions, finding the golden-ratio reciprocal ($\approx$0.62) optimal when both sources are equally sized; their result requires the contamination ratio k, unavailable in web-scale settings, so we use it as an oracle baseline. Wang et al. (2026) prove that contaminated recursive training converges at a rate governed by the effective real-data fraction α; this motivates our quality threshold, since $\alpha_{\text{eff}}(b) \geq \tau$ guarantees a convergence-rate lower bound. We operationalize this without the known-proportion assumption, estimating α_{eff} from batch-level statistics online. Score calibration (Guo et al., 2017) is relevant because miscalibration systematically biases α_{eff}; for the stability constraint, we build on Kish's effective sample size (Kish, 1965).

3 Method

Let $\mathcal{D} = \mathcal{D}_{\text{real}} \cup \mathcal{D}_{\text{synth}}$ denote a pretraining corpus with token-level contamination ratio $\rho = |\mathcal{D}_{\text{synth}}|/|\mathcal{D}|$, assumed unknown (Cheng et al., 2025). A detector assigns each sample x a synthetic probability $q(x) \in [0, 1]$; following Drayson et al. (2025), we weight each sample as $w(x) = (1 - q(x))^b$, where $b \geq 0$ is a sharpening exponent (b=0 recovers uniform training). Rather than selecting b via grid search (Drayson et al., 2025), we derive b per-batch from the batch's q-score distribution using two opposing constraints.

Given a batch $\mathcal{B} = \{x_1, \ldots, x_B\}$ with scores $q_i = q(x_i)$ and weights $w_i(b) = (1 - q_i)^b$:

Quality floor. The *effective authenticity* $\alpha_{\text{eff}}(b) = \sum_i w_i(b)(1-q_i)/\sum_i w_i(b)$ measures the weighted mean detector confidence that samples are real. We require $\alpha_{\text{eff}}(b) \geq \tau$, yielding a lower bound b_τ on the exponent, motivated by convergence theory under contamination (Wang et al., 2026). α_{eff} correlates with the true batch-level real fraction at Pearson $r \approx 0.79$ (Appendix K); limitations are discussed in §6.

Stability ceiling. Kish's effective sample size $\text{ESS}(b) = (\sum_i w_i)^2/\sum_i w_i^2$ (Kish, 1965) decreases monotonically in b (Appendix A). We require $\text{ESS}(b) \geq n_{\min}$, yielding an upper bound b_{ESS}.

Algorithm 1 Per-batch adaptive b^* selection

Require: Batch scores $\{q_i\}_{i=1}^B$, threshold τ, ESS floor $n_{\min}$, bound $b_{\max}$, tol. ε
Ensure: Optimal sharpening exponent b^*

```
if α_eff(0) ≥ τ then
    return 0 {clean batch}
end if
b_τ ← BISECT(α_eff(·)=τ, [0, b_max], ε)
if ESS(b_τ) ≥ n_min then
    return b_τ {α_eff-binding}
else
    b_ESS ← BISECT(ESS(·)=n_min, [0, b_max], ε)
    return b_ESS {ESS-binding}
end if
```

Adaptive b^*. Since α_{eff} is non-decreasing and ESS is decreasing in b, the two constraints define an interval $[b_\tau, b_{\text{ESS}}]$. The optimal exponent implements *minimum intervention*:

$$b^* = \begin{cases} 0 & \text{if } \alpha_{\text{eff}}(0) \geq \tau, \\ b_\tau & \text{if } b_\tau \leq b_{\text{ESS}}, \\ b_{\text{ESS}} & \text{otherwise (stability precedence)}, \end{cases} \tag{1}$$

applying the smallest b that achieves the quality floor. Each batch falls into one of three regimes—clean, α_{eff}-binding, or ESS-binding (Figure 2)—whose frequencies we analyze in §5.3. The resulting b^* trajectory (Figure 8, Appendix) varies substantially across batches, with all three regimes alternating throughout, confirming per-batch adaptation to local composition rather than global dynamics.

Given b^*, normalized weights $\pi_i = w_i(b^*)/\sum_j w_j(b^*)$ define the weighted loss $\mathcal{L} = \sum_i \pi_i \ell_i$; b^* is found via bisection (Algorithm 1) in $\mathcal{O}(B \log(b_{\max}/\varepsilon))$ operations, negligible beside the forward pass. Defaults: τ=0.7, $n_{\min}$=$\lfloor B/2 \rfloor$; sensitivity in §5.3.

4 Experimental Setup

Training Data. We sample a 132M-token subset of FineWeb (Penedo et al., 2024) as the real-data source, comparable to typical domain-adaptive continued pretraining budgets (Gururangan et al., 2020). To simulate web-scale synthetic contamination, we generate continuations with Qwen3-0.6B (Yang et al., 2025), the same architecture used

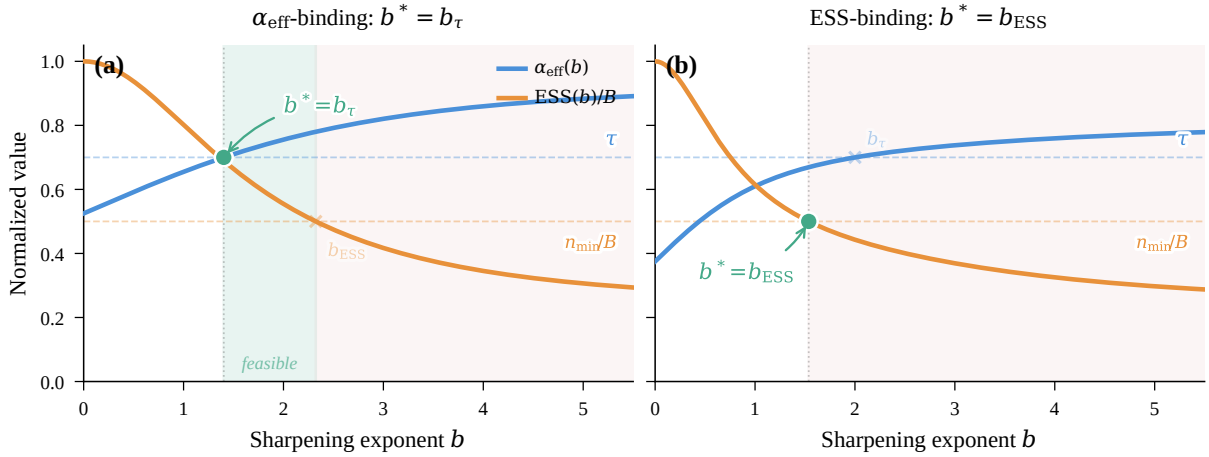


Figure 2: Dual-constraint b^* selection for two representative batches (B=4, τ=0.7, $n_{\min}$=2). **(a)** Quality constraint binds: $\alpha_{\text{eff}}(b)$ reaches τ before ESS(b) drops below $n_{\min}$, so $b^* = b_\tau$. **(b)** Stability constraint binds: ESS(b) hits the floor first, so $b^* = b_{\text{ESS}}$. Green: feasible interval; red: over-sharpening.

for continued pretraining, and inject them at token-level ratio $\rho = 0.4$ via uniform random mixing.[1] This self-contamination design means generator and trainee share the same distribution, providing a controlled testbed with known ground-truth labels but limiting ecological validity; cross-model transfer results (Appendix O) are mixed, and we discuss implications for detection difficulty in §7.

Detector. We train a TF-IDF + Logistic Regression classifier on a held-out labeled split (AUC = 0.94, calibrated ECE = 0.05 after temperature scaling with T=0.204). The resulting q-scores have mean 0.60, standard deviation 0.37, and median 0.71, with 46.3% of samples in the uncertain range $[0.2, 0.8]$. We use this lightweight detector rather than neural alternatives (e.g., ModernBERT at AUC $\geq$ 0.98 in Drayson et al. 2025) to test whether score *calibration* may matter as much as raw discrimination for reweighting. To evaluate detector sensitivity, we additionally test two generative baselines: a GPT-2-based perplexity scorer (AUC = 0.91) and an N-gram language model perplexity scorer (AUC = 0.73); results are reported in §5.2 (implementation details in Appendix M).

Training. Table 1 summarizes the training configuration. We use B=4 as the default micro-batch size, reflecting memory constraints typical of continued pretraining setups; §5.3 demonstrates that larger batches amplify the adaptive advantage. All main comparisons use three random seeds (42, 123, 456); the primary adaptive-vs-uniform comparison is extended to five seeds (adding 789, 234).

[1] ρ denotes the token-level contamination ratio. Because synthetic documents are shorter on average (304 vs. 560 words), the document-level synthetic fraction is 55.6% at ρ=0.4.

Model	Qwen3-0.6B	Optimizer	AdamW
Steps	6,103	Peak LR	5×10^{-5}
Micro-batch	B=4	Schedule	linear + warmup
Grad. accum.	4 (B_{eff}=16)	Seq. len.	2048
Precision	bf16	GPU	1× RTX PRO 6000

Table 1: Training configuration for continued pretraining.

Evaluation. We report held-out perplexity (PPL) on 5,000 documents from a FineWeb holdout set (∼4M tokens) containing only human-authored text, via sliding-window evaluation (stride 1024). All evaluations are post-hoc with a unified pipeline; training involves no inline evaluation.

Methods. We compare six approaches:

- **Uniform**: $w_j = 1$ for all j (no contamination handling).
- **Grid-b**: fixed $b \in \{0.1, 0.2, 0.3, 0.5, 1.0, 1.5, 2.0, 3.0, 5.0\}$ (nine candidates), best selected post-hoc by held-out PPL. Represents validation-based selection (Drayson et al., 2025).
- **ESS-only**: $b^* = \max\{b : \text{ESS}(b) \geq n_{\min}\}$, the maximum-intervention variant retaining only the stability constraint, ablating α_{eff}.
- **Golden Ratio**: $w = 1/\varphi \approx 0.618$ for synthetic ($q_j > 0.5$), $w = 1$ otherwise (He et al., 2025). Requires known provenance labels (oracle).
- **Adaptive** (ours): per-batch b^* from the dual-constraint formulation (§3), with $\tau = 0.7$ (de-

fault; τ=0.8 comparable, §5.3) and $n_{\min} = 2$ (half the micro-batch size).

- **Detect & Remove**: hard filtering that discards samples with $q_j > 0.5$ and trains uniformly on the remainder. Requires a calibrated detection threshold; threshold sensitivity is analyzed in §5.1.

5 Results

5.1 Main Results

Table 3 reports held-out perplexity at $\rho = 0.4$ across five seeds. Continued pretraining reduces PPL from 22.970 to the 17.3–17.8 range; method comparisons operate within a narrow residual.

Adaptive b^* achieves 5-seed mean PPL 17.674, compared to 17.697 for uniform (Δ=−0.023, directionally consistent across all seeds) and 17.620 for grid b=0.3 (3-seed mean, post-hoc oracle; gap +0.054; Table 4). A paired t-test yields $t(4)$=−8.21, p=0.0012; after Bonferroni–Holm correction for nine family-wise comparisons, p_{BH}=0.011, reaching conventional significance at α=0.05. The engineering benefit is eliminating b grid search (nine full training runs (Drayson et al., 2025), ∼16 GPU-hours at 0.6B scale, scaling linearly with model size), though grid search yields a larger improvement. The grid-best b=0.3 is a post-hoc oracle from nine candidates with 5.5× higher cross-seed standard deviation than adaptive (σ=0.033 vs. 0.006); the noisy PPL plateau at $b \in [0.1, 0.5]$ (Figure 3) makes validation-based selection unreliable. At the nearest fixed point (b=1.0), PPL equals uniform (17.692 vs. 17.694): per-batch adaptation is essential.

The ablation hierarchy at ρ=0.4 (Adaptive < Uniform < ESS-only < Random < Permuted, Table 3) reveals two requirements for effective reweighting: detector signal and minimum intervention. Random weights $\mathcal{U}(0, 2)$ (17.802) confirm that non-uniform weighting alone does not help; permuted q-scores (17.919, Δ=+0.225) further isolate provenance signal as load-bearing, since shuffling destroys the detector's sample-level information while preserving identical weight non-uniformity. ESS-only (mean 17.737±0.021) uses the same q-scores but maximizes b up to the stability ceiling, concentrating weights on a few high-confidence samples; it degrades PPL by 0.043 relative to uniform. Only the dual-constrained adaptive method, which limits b to the minimum needed, improves in-distribution PPL over uniform

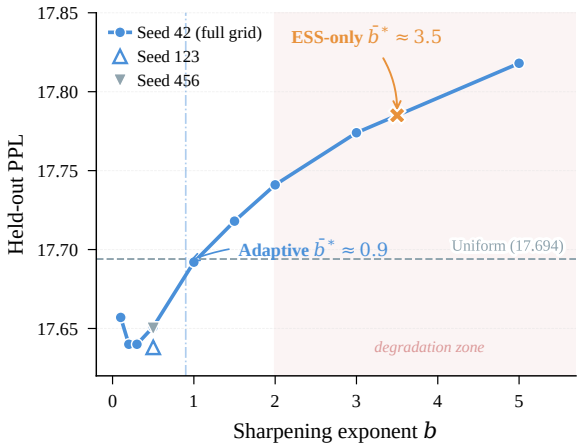


Figure 3: Held-out PPL as a function of fixed sharpening exponent b (grid search, ρ=0.4). Seed 42 provides the full nine-point sweep (circles); cross-seed markers at b=0.5 (triangles) confirm the plateau region. The shaded region marks $b \geq 2$ where PPL degrades past the uniform baseline; ESS-only b^* (overall mean ≈3.5) falls in this degradation zone, while adaptive b^* (mean ≈0.9) operates near the curve minimum.

(mean Δ=−0.023, directionally consistent across all seeds). Figure 3 contextualizes both results: adaptive b^* operates near the curve minimum while ESS-only b^* (mean ≈3.5) falls in the degradation zone.

Hard filtering and the epoch confound. Detect & Remove at its best threshold (q>0.3) achieves 3-seed mean PPL 17.250 ± 0.001 at full training length, apparently outperforming all reweighting methods. However, filtering retains only 31.5% of tokens, so the model sees surviving data ∼3.2 *content epochs* versus one for reweighting. An epoch-matched control (D&R at q>0.5, 2,533 steps, ∼1.0 content epochs) yields PPL 17.657 ± 0.005 vs. 17.674 ± 0.007 for adaptive b^* (Δ=0.017); the bulk of D&R's full-length advantage (0.394 PPL) is attributable to multi-epoch exposure (Table 2). The raw threshold–PPL curve is U-shaped (q>0.3: 17.25, q>0.9: 17.42), but epoch-matched comparisons eliminate the U-shape, revealing monotonically decreasing PPL that saturates at $q \geq 0.5$ (Figure 4; Appendix H): the apparent optimum at q>0.3 reflects maximally compensating multi-epoch exposure, not filtering quality. For practitioners, this means D&R's published advantages should be interpreted with caution: in continued pretraining scenarios where the corpus is small relative to training compute, aggressive filtering can inadvertently create multi-epoch regimes that inflate

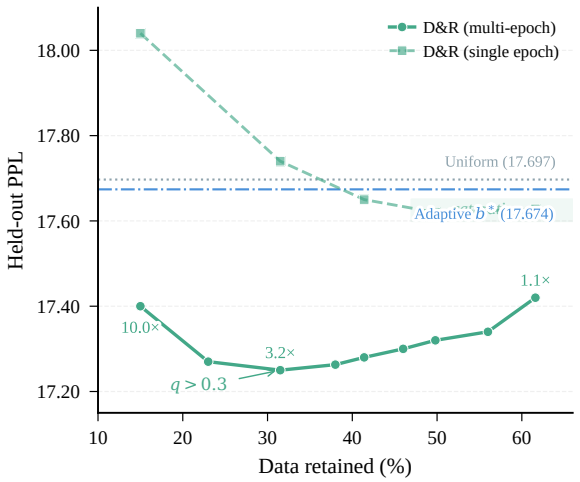


Figure 4: D&R threshold–PPL relationship (ρ=0.4, 3 seeds). Blue: raw (multi-epoch) PPL with effective epoch counts annotated. Orange: epoch-matched PPL (single content epoch), saturating at the adaptive b^* level (green dashed).

Method	Epochs	PPL (raw) ↓	PPL (ep.-matched) ↓
D&R (q>0.3)	∼3.2	**17.250** ± .001	17.74
D&R (q>0.5)	∼2.4	17.280 ± .010	17.657 ± .005
D&R (q>0.7)	∼2.0	17.32	17.62
Adaptive b^*	∼1.0	17.674 ± .007	
Uniform	∼1.0	17.697 ± .008	

Table 2: D&R threshold comparison (ρ=0.4). Raw PPL traces a U-shape across thresholds; epoch-matched PPL (single content epoch) decreases monotonically and saturates near the adaptive b^* level. D&R's 0.394 PPL advantage (at q>0.5) shrinks to Δ=0.017 when content epochs are equalized. Multi-seed (≥3) where shown; q>0.3 and q>0.7 epoch-matched are single-seed. Full 9-threshold sweep in Table 9.

apparent gains. Controlled comparison requires either epoch-matching or, at minimum, reporting the effective content-epoch count alongside PPL.

5.2 Robustness and Boundary Conditions

Across three seeds at $\rho = 0.6$, adaptive b^* (17.731 ± 0.004) outperforms uniform on in-distribution PPL (17.763 ± 0.008, Δ=−0.032, 3 seeds, paired t=−9.95, p_{BH}=0.064), exceeding the ρ=0.4 improvement (Δ=−0.023). The ablation hierarchy is preserved (Grid 17.634[2] < Adaptive 17.731 < Uniform 17.763 ≈ ESS-only 17.763 < Random 17.861), but the ESS-only/uniform gap collapses to effectively zero (three-seed means both 17.763) as ESS binding rises to ∼49%. Random remains substantially worse than uniform (gap 0.098, σ=0.025): detector signal is load-bearing even when the stability constraint dominates; the adaptive method's continued advantage is driven by the α_{eff} quality constraint, which operates independently of the ESS ceiling. Three-seed paired comparison confirms τ=0.7 outperforms τ=0.8 on both holdout and out-of-distribution (OOD) metrics (§5.3).

[2]Single seed; full per-seed results in Appendix J.

Method	Seed 42	Seed 123	Seed 456	Mean
Uniform	17.694	17.686	17.702	17.697 ± .008
Adaptive b^* (ours)	**17.680**	**17.666**	**17.677**	**17.674** ± .006
ESS-only	17.730	17.720	17.760	17.737 ± .021
Random $\mathcal{U}(0, 2)$	17.830	17.800	17.777	17.802 ± .027
Permuted q-scores	17.919	–	–	17.919

Table 3: Main comparison: held-out PPL (↓) at ρ=0.4, 1 content epoch. Adaptive and Uniform: 5-seed means (seeds 42, 123, 456, 789, 234); ESS-only, Random, and Permuted: seed 42 (Permuted single seed). Adaptive uses τ=0.7, $n_{\min}$=2. Permuted randomly shuffles q-scores across samples, destroying provenance signal while preserving weight non-uniformity (Δ=+0.225).

Method	Seeds	Oracle?	Epochs	Mean PPL ↓
Grid (b=0.3)	3	post-hoc	1	**17.620** ± .033
Golden Ratio	1	known ρ	1	17.654
D&R (q>0.3)	3	post-hoc	∼3.2	**17.250** ± .001

Table 4: Auxiliary baselines at ρ=0.4. Grid-best selected post-hoc from nine candidates (Appendix C). Golden Ratio requires known ρ (He et al., 2025). D&R uses ∼3.2 content epochs (Table 2).

Scale. At 1.7B parameters (Qwen3-1.7B, ρ=0.4, τ=0.8, 3 seeds), the method scales without degradation but the benefit diminishes. Adaptive b^* achieves holdout PPL 13.26 vs. uniform 13.27 (Δ=−0.013); OOD results are mixed in direction across domains (OWT Δ=−0.008, Wiki Δ=+0.013) and downstream accuracy is unchanged (Δ≈+0.002). The diminished effect is consistent with larger models being more robust to self-generated contamination: Qwen3-1.7B's own outputs are likely closer to its training distribution, reducing the distributional shift that reweighting corrects. A preliminary Llama-3.2-3B cross-model result (Appendix O) shows mixed downstream effects (MMLU +1.6 percentage points, HellaSwag −0.6 percentage points), leaving cross-architecture generalization as an open question.

Out-of-distribution evaluation. To test whether the in-distribution improvement reflects distribution matching, we evaluate all three from-scratch

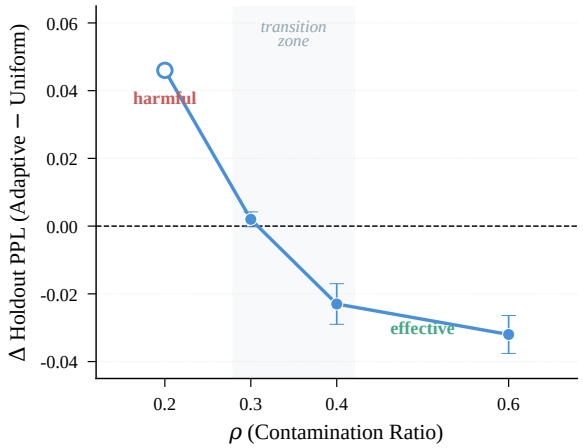


Figure 5: Effect of contamination ratio on adaptive reweighting effectiveness. Δ = adaptive − uniform holdout PPL (lower is better). Error bars show ± 1 SD across seeds. The method transitions from harmful to beneficial between ρ=0.3 and 0.4 (shaded).

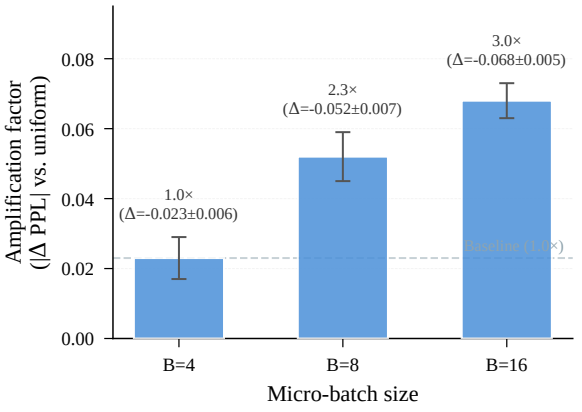


Figure 6: Batch-size amplification of adaptive reweighting ($n_{\min}=\lfloor B/2 \rfloor$). The adaptive advantage over uniform increases monotonically with micro-batch size, from 1.0× (B=4) to 3.0× (B=16); Δ values and ± 1 SD shown above each bar.

seeds on OpenWebText and Wikipedia. At ρ=0.4, adaptive reweighting improves mean PPL on all three evaluation domains: FineWeb $\Delta=-0.023$ (5 seeds), OpenWebText $\Delta=-0.028$, Wikipedia $\Delta=-0.029$ (3 seeds each); downstream accuracy is identical (0.5414). OOD improvements exceed the in-distribution gain, indicating genuine quality improvement rather than distribution matching toward the training mixture (τ=0.8's OOD effect is smaller; τ=0.7's advantage reflects active OOD improvement, §5.3).

Calibration, not just discrimination. Replacing TF-IDF+LR with generative perplexity detectors reverses the advantage across three seeds (GPT-2: AUC = 0.91, ECE = 0.131, $\Delta=+0.098$; N-gram: AUC = 0.73, ECE = 0.069, $\Delta=+0.129$; Figure 1b; Appendix L). The TF-IDF vs. GPT-2 contrast (comparable AUC, 2.7× ECE gap) isolates calibration as an empirically observed requirement in our self-contamination setting; N-gram's better ECE (0.069) does not compensate for insufficient AUC (0.73), confirming that both discrimination and calibration are required (Figure 1b).

5.3 Hyperparameter Sensitivity

Quality threshold τ. τ=0.7 outperforms τ=0.8 on holdout at both contamination levels (ρ=0.4: 17.674 vs. 17.683; ρ=0.6: paired $t=-4.70$, $p<0.05$; §5.2) and achieves larger OOD gains (ρ=0.4: $\Delta=-0.028/-0.029$ vs. $-0.005/-0.009$; Table 7, Appendix G). $\tau \in [0.5, 0.8]$ all improve holdout over uniform, while $\tau>0.8$ degrades (Appendix D). We also tested per-batch adaptive τ clamping, which did not improve over fixed thresholds (Appendix E).

ESS floor $n_{\min}$. The default $n_{\min}$=2 ($n_{\min}/B$=0.5) is optimal; extremes degrade ($n_{\min}$=1: PPL 17.742±0.009, 3 seeds, worse than uniform; $n_{\min}=B$: recovers uniform). Neither constraint alone improves over uniform ($n_{\min}$=1: +0.048; ESS-only: +0.043), confirming the dual constraint is necessary. The coupling $n_{\min}=\lfloor B/2 \rfloor$ is also empirically motivated: at B=8, keeping $n_{\min}$=2 (ratio 0.25) degrades PPL by +0.036 vs. uniform, whereas $n_{\min}$=4 (ratio 0.5) yields $\Delta=-0.052\pm0.007$. Under this coupling, the adaptive advantage monotonically increases across $B \in \{4, 8, 16\}$: $\Delta=-0.023\pm0.006$ (5 seeds), -0.052 ± 0.007 (3 seeds, p=0.006), and -0.068 ± 0.005 (3 seeds), corresponding to 1.0×, 2.3×, and 3.0× amplification ($p<0.01$ at $B \geq 8$). The gap to grid-best narrows from 0.040 (B=4) to 0.009 (B=16, seed 42; Figure 6).

Constraint regime. Over 24,412 micro-batches (seed 42), 5.8% are clean ($\alpha_{\text{eff}}(0) \geq \tau$, b^*=0), ∼50% are α_{eff}-binding (quality constraint dominates), and ∼44% are ESS-binding (stability constraint dominates); the distribution is stable across seeds (coefficient of variation <2%; Figure 7). The α_{eff}-binding majority indicates that detector quality signal—not training stability—is the primary bottleneck, reinforcing the calibration requirement (§5.2).

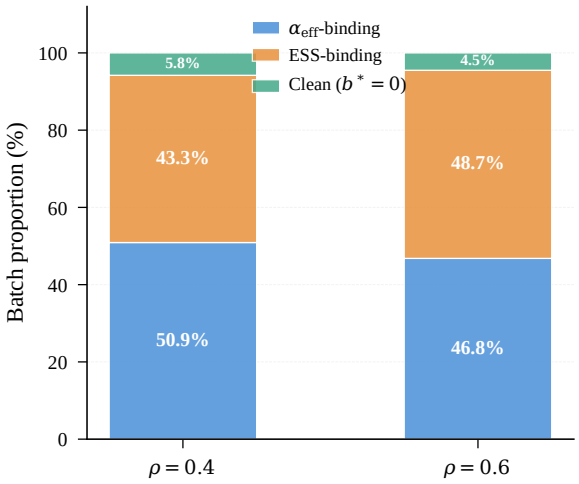


Figure 7: Constraint regime decomposition at two contamination levels (seed 42). At ρ=0.4, ∼50% of batches are α_{eff}-binding and ∼44% ESS-binding, with ∼6% clean; at ρ=0.6, the constraints are nearly balanced (∼49% ESS-binding, ∼47% α_{eff}-binding).

Condition	Req'd?	Evidence	Δ PPL
Detector AUC ≥ 0.9	Yes	N-gram (0.73) degrades	+0.129
Detector ECE ≤ 0.05	Yes	GPT-2 (0.131) degrades	+0.098
Contamination $\rho \geq 0.4$	Yes	ρ=0.2 degrades	+0.046
Generator $\neq$ trainee	Rec.	1.7B: Δ=−0.013, OOD mixed	—

Table 5: Decision summary for practitioners. Δ values are relative to uniform (positive = worse). When any required condition is violated, uniform is the safer default.

6 Discussion

The epoch confound in D&R evaluation. The epoch confound identified in §5.1 accounts for >95% of D&R's reported advantage; the residual (Δ=0.017, 3 seeds) falls within cross-seed variance and is robust to LR schedule choice (Δ=0.006 under independent schedules). This confound was invisible in Drayson et al. (2025) because from-scratch pretraining on large corpora rarely completes even one epoch; in continued pretraining, aggressive D&R filtering (retaining 31.5% at q>0.3) creates ∼3.2 content epochs, making the confound quantitatively significant. A permuted-score control (+0.225 vs. uniform) confirms that detector signal is load-bearing.

When uniform beats reweighting. Table 5 summarizes three conditions, each necessary for effective reweighting: (i) sufficient detector discrimination *and* calibration (AUC $\geq$ 0.9, ECE $\leq$ 0.05; §5.2); (ii) contamination $\rho \geq 0.4$, below which the method degrades (+0.046 at ρ=0.2) with a sharp transition in $[0.3, 0.4]$ (Figure 5, Appendix Q); (iii) generator–trainee mismatch (at 1.7B the benefit diminishes to Δ=−0.013 with OOD mixed). When any condition is violated, uniform is the safer default.

Effect size and downstream impact. The in-distribution improvement (Δ=−0.023 at ρ=0.4) does not transfer to downstream accuracy (0.5414 vs. 0.5414, 5 seeds; Appendix I), consistent with nonlinear PPL-to-accuracy transfer (Tay et al., 2023; Ye et al., 2025); the paradigm's impact at current scales is modest (Appendix R). Table 17 (Appendix) summarizes additional results.

Limitations of α_{eff} as a proxy. The α_{eff} diagnostic correlates with the true batch-level real fraction at Pearson $r \approx 0.79$ (∼63% of variance; Appendix K), stable across seeds (Spearman $\rho_s \geq 0.86$; Table 13). The ceiling is set by detector calibration, not batch size: scaling to B=16 leaves R^2 in the 0.59–0.63 range, though the adaptive advantage still increases monotonically (1.0× to 3.0×; §5.3). Because the convergence guarantee of Wang et al. (2026) requires the *true* real fraction, the ∼37% unexplained variance means some batches satisfying $\alpha_{\text{eff}} \geq \tau$ may have true fractions below τ; the $n_{\min}$ constraint mitigates this by capping b at the ESS floor, preventing weight concentration where the proxy is unreliable (Owen, 2013).

7 Conclusion

We identify three effectiveness conditions for detect-and-reweight—epoch-matched evaluation, detector calibration alongside discrimination, and minimum intervention over unconstrained maximization—and demonstrate the third via a dual-constrained adaptive b^* that improves over uniform training at $\rho \geq 0.4$ without per-model tuning, though without measurable downstream gains at 0.6B scale. For practitioners, reweighting is worthwhile only when $\rho \geq 0.4$ and the detector is well-calibrated (Table 5).

Our epoch-matching analysis reveals that >95% of D&R's reported advantage is attributable to multi-epoch exposure rather than filtering quality, cautioning against interpreting raw PPL comparisons in data-limited continued pretraining. The permuted-score ablation (Δ=+0.225) confirms that detector provenance signal—not mere weight non-uniformity—drives the improvement. Looking forward, extending this framework to full pretrain-

ing scale, multi-source web contamination, and broader detector families (e.g., fine-tuned neural classifiers) remains an open direction.

Limitations

- **Scope.** All experiments use Qwen3 (0.6B and 1.7B) under self-contamination with 132–200M tokens of continued pretraining. Generalization to other architectures, full pretraining scale (>100B tokens), or realistic multi-source web contamination—where generator–corpus distribution mismatch may reduce detector accuracy—remains untested.
- **Detector coverage.** Only three detectors from two families (discriminative TF-IDF+LR and generative perplexity scorers) were evaluated; fine-tuned neural classifiers (e.g., ModernBERT) and zero-shot methods (e.g., DetectGPT, Binoculars) remain untested, so the AUC/ECE thresholds reported here may shift with broader detector coverage.

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A Proofs

Proposition 1 (α_{eff} monotonicity)**.** *$\alpha_{\text{eff}}(b)$ is non-decreasing in b, and strictly increasing whenever the batch contains at least two samples with distinct q-values.*

Proof of Proposition 1. Let $w_i(b) = (1-q_i)^b$, $W(b) = \sum_{j=1}^{B} w_j(b)$, and $\pi_i(b) = w_i(b)/W(b)$. Then

$$\alpha_{\text{eff}}(b) = \sum_{i=1}^{B} \pi_i(b)\,(1-q_i).$$

Taking the derivative with respect to b:

$$\begin{aligned}\frac{d\pi_i}{db} &= \pi_i\left(\frac{d\ln w_i}{db} - \frac{d\ln W}{db}\right)\\ &= \pi_i\Big(\ln(1-q_i) - \textstyle\sum_j \pi_j \ln(1-q_j)\Big)\,.\end{aligned}$$

Substituting,

$$\begin{aligned}\frac{d\alpha_{\text{eff}}}{db} &= \sum_i (1-q_i)\,\pi_i(\ln(1-q_i) - \mathbb{E}_\pi[\ln(1-q)])\\ &= \mathrm{Cov}_\pi\big(1-q_i,\, \ln(1-q_i)\big).\end{aligned}$$

Since $(1-q_i)$ and $\ln(1-q_i)$ are both increasing functions of $(1-q_i)$, they are positively associated, so the covariance is non-negative. □

Proposition 2 (ESS monotonicity)**.** *$\mathrm{ESS}(b)$ is non-increasing in b, and strictly decreasing whenever the batch contains at least two samples with distinct q-values.*

Proof. Write $\mathrm{ESS}(b) = 1/\sum_i \pi_i(b)^2$, where $\pi_i(b) = w_i(b)/W(b)$. The denominator $\sum_i \pi_i^2$ is the Herfindahl index H. Differentiating H with respect to b:

$$\frac{dH}{db} = 2\sum_i \pi_i^2\big(\ln(1-q_i) - \mathbb{E}_\pi[\ln(1-q)]\big).$$

Since $\pi_i \propto (1-q_i)^b$, larger π_i implies larger $\ln(1-q_i)$; the π^2-weighted sum assigns greater mass to terms where $\ln(1-q_i) > \mathbb{E}_\pi[\ln(1-q)]$, so $dH/db \geq 0$, with strict inequality when at least two q-values differ. Thus H is non-decreasing and $\mathrm{ESS} = 1/H$ is non-increasing. This is a standard property of exponential tilting (Kish, 1965; Owen, 2013). □

B Algorithm Details

See Algorithm 1 (§3) for the pseudocode.

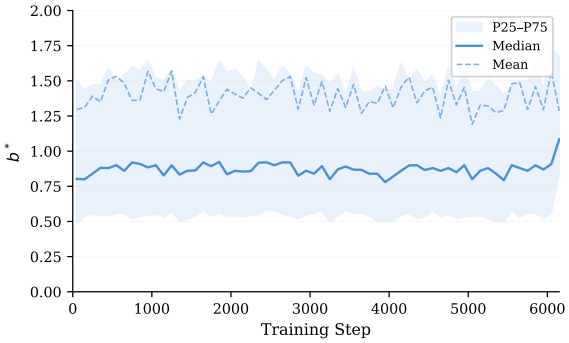


Figure 8: Adaptive b^* trajectory over 6,103 training steps (seed 42, ρ=0.4). The exponent alternates between clean (b^*=0), α_{eff}-binding, and ESS-binding regimes with no systematic drift, reflecting per-batch adaptation to local q-score composition.

We compute b^* at the micro-batch level (B=4) rather than the effective batch (B=16 after gradient accumulation) because q-score statistics exhibit

sufficient variance within micro-batches: Spearman r=0.699 with the true real fraction, and at B=16 the adaptive advantage amplifies 3.0× (Δ=−0.068±0.005; §5.3).

C Grid Search Details

Table 6 reports held-out PPL for all grid-b values at $\rho = 0.4$ (seed 42).

b	PPL ↓
0.1	17.657
0.2	17.640
0.3	**17.640**
0.5	17.651
1.0	17.692
1.5	17.718
2.0	17.741
3.0	17.774
5.0	17.818

Table 6: Full grid search results (ρ=0.4, seed 42). $b \in [0.1, 0.5]$ forms a broad plateau (three-seed mean for b=0.1: 17.655 ± 0.009); b=0.3 is the post-hoc best.

D Quality Threshold Sweep

τ	Holdout ↓	Δ_H	OWT ↓	Δ_O	Wiki ↓	Δ_W
0.50	17.688	−0.006	–	–	–	–
0.60	17.681	−0.013	–	–	–	–
0.70*	**17.674**±.006	**−0.020**	**18.347**±.003	**−0.028**	**15.330**±.015	**−0.029**
0.80*	17.683±.006	−0.011	18.370±.005	−0.005	15.350	−0.009
0.90	17.712	+0.018	–	–	–	–
0.95	17.723	+0.029	–	–	–	–

Table 7: Quality threshold sweep ($n_{\min}$=2, ρ=0.4). Δ relative to 3-seed uniform baseline (Holdout 17.694, OWT 18.375, Wiki 15.359). * τ=0.7: 5-seed mean (SD from 5 seeds); τ=0.8: 3 from-scratch seeds (Appendix P). The 5-seed comparison in Table 3 yields Δ=−0.023. Other values from single-seed exploratory sweep (seed 42, holdout only). τ=0.7 achieves the lowest holdout and largest OOD gains ($|\Delta| \geq 0.028$); τ=0.8's OOD effect is smaller ($|\Delta| \leq 0.009$).

E Batch-Adaptive τ

We tested a per-batch adaptive variant that clamps τ between 0.5 and the fixed threshold based on local $\alpha_{\text{eff}}(0)$, allowing the quality floor to relax in batches with low estimated real fraction. At both contamination levels ($\rho \in \{0.4, 0.6\}$), the adaptive variant did not outperform the fixed threshold (single-seed exploratory at each level), suggesting the quality constraint does not require adaptation to local contamination density. This supports the "two hyperparameters with good defaults" characterization: neither τ nor $n_{\min}$ benefits from per-batch tuning.

F Constraint Regime Decomposition

See Figure 7 (§5.3) for the constraint regime visualization. Cross-seed stability: the regime proportions are stable across seeds (CV <2%), confirming that the decomposition is a property of the data distribution rather than initialization noise.

G τ–ρ Rank Preservation

Figure 9 shows the τ–PPL relationship at two contamination levels.

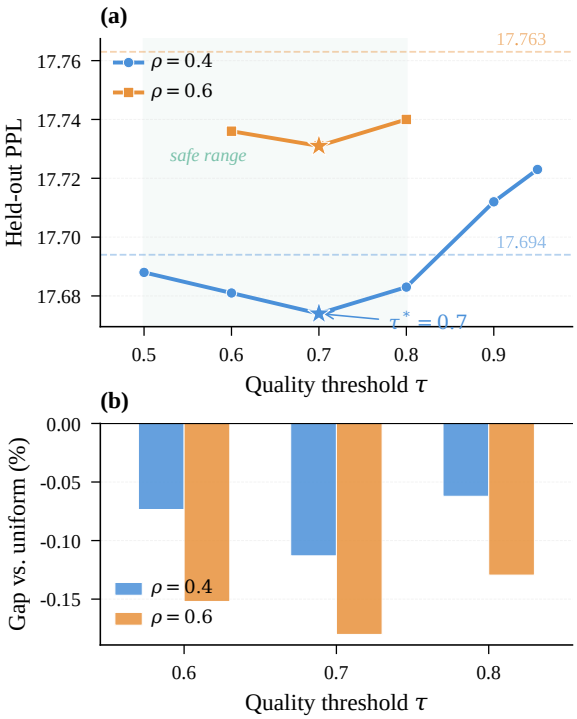


Figure 9: (a) Held-out PPL as a function of quality threshold τ at $\rho \in \{0.4, 0.6\}$. τ=0.7 and τ=0.8 yield comparable holdout PPL at ρ=0.4; at ρ=0.6, τ=0.7 is statistically better (paired t=−4.70, p<0.05, 3 seeds, Δ=−0.005). τ=0.7 also achieves lower OOD perplexity at both levels (ρ=0.6: Δ_{OWT}=−0.014, Δ_{Wiki}=−0.013, 2 seeds). (b) Normalized PPL gap relative to uniform at shared τ values.

H Hybrid and Threshold Comparison

Table 8 compares method families across contamination levels. Table 9 shows D&R sensitivity to the detection threshold at both full training length and epoch-matched settings. Raw PPL traces a U-shaped curve with a minimum at q>0.3 (17.250, 3 seeds), but this curve conflates filtering quality with

multi-epoch exposure: aggressive thresholds retain less data and repeat it more often. Epoch-matched comparisons (rightmost column of Table 9) remove the multi-epoch confound: the U-shape vanishes and PPL decreases monotonically, saturating near the adaptive b^* level at $q \geq 0.5$. Figure 4 visualizes the two curves.

Method	ρ=0.4	ρ=0.6
Uniform	17.694	17.766
Adaptive b^*	17.689	17.740
Hybrid (q>0.95 + adaptive)	17.514	17.530*
Detect & Remove (q>0.5)	17.285	–
Detect & Remove (q>0.3)	17.250	–

Table 8: Method comparison across contamination levels (seed 42, τ=0.8). *Hybrid at ρ=0.6 filters high-confidence synthetic (q>0.95), reducing effective ρ to $\approx$0.4; PPL is not directly comparable to ρ=0.6 baselines.

Thresh.	**Ret. %**	**Epochs**	**Raw PPL** ↓	**Ep.-matched** ↓
$q > 0.1$	15.0%	∼10.0	17.40	18.04
$q > 0.2$	∼23%	∼5.0	17.27	–
$q > 0.3$	31.5%	∼3.2	**17.250**†	17.74
$q > 0.4$	∼38%	∼2.8	17.26	–
$q > 0.5$	41.4%	∼2.4	17.28	17.65
$q > 0.6$	∼46%	∼1.7	17.30	–
$q > 0.7$	49.8%	∼2.0	17.32	17.62
$q > 0.8$	∼56%	∼1.3	17.34	–
$q > 0.9$	61.6%	∼1.1	17.42	17.628

Table 9: D&R threshold sensitivity (ρ=0.4, seed 42 unless noted). Raw PPL traces a U-shaped curve (minimum at q>0.3); epoch-matched PPL (single content epoch) decreases monotonically and saturates near the adaptive b^* level ($\approx$17.65) at $q \geq 0.5$, confirming the U-shape is a multi-epoch artifact. † 3-seed mean 17.250 ± 0.001.

See Figure 4 (§5.1) for the dual-curve visualization.

I Downstream Task Evaluation

Table 10 reports downstream benchmark accuracy for adaptive vs. uniform training at ρ=0.4 (seed 42, Qwen3-0.6B). Across multiple seeds, downstream differences are mixed in direction and uniformly within standard error, confirming that adaptive reweighting has no meaningful downstream impact.

J Higher Contamination Details

Table 11 reports results at ρ=0.6. The full five-method ablation preserves the hierarchy from ρ=0.4 across all three seeds: Grid < Adaptive < Uniform ≈ ESS-only < Random.

Benchmark	**Metric**	**Adaptive**	**Uniform**	**Δ**
Seed 42 (checkpoint resume)				
ARC-Challenge	acc_norm	0.343	0.348	−0.005
ARC-Easy	acc	0.650	0.657	−0.007
HellaSwag	acc_norm	0.479	0.484	−0.005
PIQA	acc	0.687	0.686	+0.001
WinoGrande	acc	0.578	0.586	−0.008
From-scratch 2-seed mean (seeds 123, 456)				
Mean Δ across 5 tasks				+0.001
Direction				mixed (seed 123: 3/5−; seed 456: 3/5+)

Table 10: Downstream benchmarks (Qwen3-0.6B, ρ=0.4). Seed 42 used a checkpoint-resumed adaptive model; from-scratch seeds (123, 456) show mixed direction with mean $|\Delta| \leq 0.01$. All differences within standard error ($\pm$0.01). Models use τ=0.8; τ=0.7 yields identical 3-seed means (0.5414).

Method	**Seed 42**	**Seed 123**	**Seed 456**	**Mean**
Grid (b=0.3)‡	**17.634**	–	–	–
Adaptive b^*	17.735	17.727	17.732	17.731 ± .004
Uniform	17.766	17.754	17.770	17.763 ± .008
ESS-only	17.770	17.760	17.760	17.763 ± .006
Random $\mathcal{U}(0, 2)$	17.832	17.870	17.880	17.861 ± .025

Table 11: Results at ρ=0.6 (Qwen3-0.6B, $n_{\min}$=2). Adaptive uses τ=0.7 (consistent with main results); other methods are τ-independent. ‡Post-hoc oracle; seed 42 only. Adaptive vs. Uniform: Δ=−0.032, paired t=−15.4, p=0.004 (full precision; 3dp rounding yields t=−9.95, p=0.010). ESS-only collapses onto Uniform (both 17.763); Random (17.861) confirms detector signal is load-bearing.

The ESS-only/Uniform gap collapses from 0.043 at ρ=0.4 to 0.000 at ρ=0.6 (three-seed means: both 17.763); ESS binding rises to ∼49% of batches (from ∼43% at ρ=0.4), and the constraints become nearly balanced rather than α_{eff}-dominant. The adaptive method maintains its advantage (Δ=−0.032, consistent across seeds) through the α_{eff} quality constraint, which operates independently of the ESS ceiling. Random reweighting (17.861 ± 0.025) is substantially worse than Uniform, with the highest cross-seed variance, confirming that uninformed weights add noise without corrective signal. The Adaptive-vs-Grid gap is 0.054 at ρ=0.4 (mean-to-mean, consistent with Table 4) and 0.097 at ρ=0.6 (adaptive 3-seed mean vs. grid seed 42; the ρ=0.6 grid result is preliminary and may not generalize across seeds).

K α_{eff} Proxy Validation

Table 12 reports the correlation between $\alpha_{\text{eff}}(0)$ (the detector-based proxy at b=0) and the true batch-level real fraction α_{true}, computed over all

training batches at ρ=0.4 (seed 42).

B	Pearson r	Spearman r	MAE↓	Bias
4	0.706	0.699	0.114	−0.029
8	0.697	0.683	0.097	−0.027
16	0.707	0.691	0.075	−0.031

Table 12: Correlation between $\alpha_{\text{eff}}(0)$ and true batch-level real fraction across batch sizes (ρ=0.4, seed 42). Values use a binarized proxy $\hat{r}_i = \mathbf{1}[q_i \leq 0.5]$ for discrete comparison; the continuous formulation used during training (§3) achieves $r \approx 0.77$–0.79 depending on batch size. The consistent negative bias (≈ -0.03) indicates that α_{eff} slightly underestimates the true real fraction, biasing the method toward conservative intervention.

Mean absolute error (MAE) decreases with batch size (0.114 → 0.075) as expected from averaging, but the rank-order information that drives b^* selection is already well-captured at B=4.

Cross-seed stability. Table 13 reports the correlation between α_{eff} and the batch mean q-score ($\bar{q}$, computed from known labels) across all three seeds.

Seed	Pearson $r(\alpha_{\text{eff}}, \bar{q})$	Spearman ρ_s	b^*=0 (%)
42	0.8019	0.8761	5.8
123	0.7994	0.8692	5.7
456	0.8006	0.8749	5.9
Spread	0.0025	0.0069	0.2

Table 13: Per-seed α_{eff}–$\bar{q}$ correlation (ρ=0.4, B=4). Cross-seed spread is <0.003 for Pearson r and <0.007 for Spearman ρ_s, confirming that the proxy's fidelity is stable across random seeds. $r(\alpha_{\text{eff}}, \bar{q}) = 0.80$ measures the proxy's fidelity to batch composition; the ground-truth correlation $r(\alpha_{\text{eff}}, \text{true real fraction}) \approx 0.79$ (§6) is slightly lower due to detector noise. The per-seed table demonstrates cross-seed stability rather than superseding the ground-truth estimate.

L Detector Comparison

See Figure 1b (§5.2) for the detector requirements visualization.

Table 14 compares three detector families for adaptive reweighting. The discriminative TF-IDF+LR detector (ECE = 0.048) produces the only improvement over uniform. The TF-IDF vs. GPT-2 contrast (comparable AUC: 0.94 vs. 0.91, but 2.7× ECE gap: 0.048 vs. 0.131) isolates calibration as an empirically observed requirement; however, N-gram's better ECE (0.069) does not compensate for its low AUC (0.73), confirming that both sufficient discrimination and good calibration are required in our setting. TF-IDF+LR's q-scores have reliable probabilistic semantics, so the α_{eff} proxy accurately tracks batch quality. Generative detectors assign high q-scores to fluent, predictable human text (false positives concentrated among high-quality samples), systematically inflating α_{eff} for clean batches and deflating it for contaminated ones, which inverts the minimum-intervention logic.

Detector	**Type**	**AUC**	**ECE**	**Mean PPL**	**Δ**
TF-IDF+LR	Discriminative	0.94	0.048	17.650$_{\pm.010}$	−0.044
GPT-2 PPL	Generative	0.91	0.131	17.792$_{\pm.005}$	+0.098
N-gram PPL	Generative	0.73	0.069	17.823$_{\pm.003}$	+0.129

Table 14: Detector comparison for adaptive reweighting (ρ=0.4, τ=0.8, 3 seeds). Uniform baseline PPL = 17.694. Δ = Adaptive − Uniform (negative is better). TF-IDF+LR calibrated via temperature scaling. The τ confound is conservative: TF-IDF+LR at τ=0.7 achieves 17.674 (Table 3).

M Detector and Synthetic Data Details

TF-IDF+LR detector. TF-IDF features (unigrams + bigrams, max 50,000 features, sublinear TF) fed to Logistic Regression (C = 1.0, L2 penalty, solver = lbfgs, max_iter = 100). Trained on a held-out labeled split (10K documents per class). Post-hoc temperature scaling applied when uncalibrated ECE exceeds 0.05: a scalar T is searched via `scipy.optimize.minimize_scalar` over $[0.1, 10.0]$ on a 20% stratified validation split, yielding T=0.204 (ECE reduced from 0.12 to 0.05). The detector is selected automatically from six configurations as the first satisfying AUC $\in [0.85, 0.98]$ and mid-range score ratio $\geq 5\%$.

Synthetic data generation. Continuations generated with Qwen3-0.6B (self-generation) using nucleus sampling (p=0.95, T=0.8), producing up to 384 tokens per continuation from a 128-token prefix prompt (512 total), targeting ∼100M synthetic tokens. Source prompts are the first 128 tokens of randomly sampled FineWeb documents, used as prefix continuations. Injected at token-level ratio ρ via uniform random mixing across the training corpus.

N Low Contamination Analysis

At ρ=0.2, the adaptive method degrades by +0.046 PPL relative to uniform (0.26% relative, seed 42). The detector's mean q-score at this contamination

level is 0.393 (vs. 0.487 at ρ=0.4), indicating that many clean samples receive moderate q-scores, blurring the provenance signal. The regime decomposition reveals 13.7% clean batches (b^*=0), 60.1% α_{eff}-binding, and 26.3% ESS-binding, with mean b^*=1.735 in reweighted batches. The high α_{eff}-binding rate (60.1% vs. ∼50% at ρ=0.4) means the method actively intervenes rather than defaulting to b^*=0, but the unreliable signal causes it to suppress useful training data. This is consistent with the generative detector failures: when q-scores do not reliably separate provenance, the minimum intervention framework amplifies the error rather than correcting it. The effective range of the method is $\rho \geq 0.4$ with $\tau \in [0.7, 0.8]$ (§5.3); lower τ values may extend the range but have not been tested at ρ=0.2.

O Cross-Model Transfer

Table 15 reports a single exploratory run with Llama-3.2-3B as a cross-generator (synthetic data generated by Llama, training on Qwen3-0.6B); this does not establish generality. Results are mixed: MMLU improves while HellaSwag slightly degrades, consistent with the pattern observed in Qwen3-1.7B downstream evaluation (§5.2). Perplexity comparison omitted due to inconsistent evaluation configurations; checkpoints were not retained for re-evaluation.

Metric	Uniform	Adaptive	Δ
MMLU (5-shot)	51.85% ± .41	53.42% ± .41	+1.57pp
HellaSwag (10-shot, acc_norm)	73.08% ± .44	72.49% ± .45	−0.60pp

Table 15: Cross-model transfer: Llama-3.2-3B generator, Qwen3-0.6B trainer (ρ=0.4, single seed). Mixed results reflect evaluation-set distribution sensitivity rather than consistent quality differences.

P Checkpoint Resume Artifact

The checkpoint-resume artifact described here affected early τ=0.8 experiments only. All results in Tables 3–2 use independent from-scratch training (τ=0.7; Table 3 Adaptive/Uniform: five seeds, others: three seeds).

The adaptive model for seed 42 in early τ=0.8 runs was trained with a checkpoint-resumed run whose first ∼1,000 steps used different hyperparameters (learning rate warmup, gradient accumulation configuration) than seeds 123 and 456, which were trained from scratch. The resume artifact inflates the in-distribution effect (Δ=−0.041 resume vs. Δ=−0.005 from-scratch at τ=0.8) and reverses the out-of-distribution direction:

Domain	Resume	From-scratch	Dir.
FineWeb (in-dist)	−0.041	−0.005	same
OpenWebText (OOD)	+0.058	−0.005	reversed
Wikipedia (OOD)	+0.137	−0.009	reversed

Δ relative to from-scratch uniform (Table 7). Resume: seed 42 original checkpoint. From-scratch: clean replication (D110).

The resume model showed apparent distribution matching (OOD degradation), which was the basis for the "distribution matching" finding in earlier analyses. From-scratch retraining eliminates this pattern: all three evaluation domains show small, directionally consistent improvements, providing no evidence of distribution matching. This underscores the importance of controlling for training artifacts when interpreting small effect sizes across evaluation domains.

Q Additional Contamination Rate

To probe the boundary between the degradation regime (ρ=0.2, Δ=+0.046) and the improvement regime (ρ=0.4, Δ=−0.023), we run adaptive b^* and uniform baselines at ρ=0.3 using the same experimental setup as the main experiments (Qwen3-0.6B, 132M FineWeb tokens, τ=0.7, $n_{\min}$=2).

Method	Seed	FineWeb ↓	OWT ↓	Wiki ↓
Uniform	42	17.422	18.113	15.204
Adaptive b^*	42	17.422	18.123	15.172
Uniform	123	17.433	18.126	15.204
Adaptive b^*	123	17.435	18.148	15.165
Uniform	456	17.419	–	–
Adaptive b^*	456	17.424	–	–
Δ (3-seed mean)	–	+0.002	–	–

Table 16: Results at ρ=0.3 (Qwen3-0.6B, 3 seeds). Holdout Δ=+0.002 ($t(2)$=1.60, p=0.25, non-significant); OOD available for seeds 42 and 123 only, mixed in direction.

The holdout difference is effectively zero (Δ=+0.002, $t(2)$=1.60, p=0.25, non-significant), placing the transition boundary between the degradation regime (ρ=0.2, Δ=+0.046) and the improvement regime (ρ=0.4, Δ=−0.023) within $[0.3, 0.4]$ (Figure 5).

At ρ=0.3, 28% of micro-batches have b^*=0 (clean, no intervention), compared to ∼6% at ρ=0.4; the low intervention frequency explains why the method neither helps nor harms. The

progression across contamination levels—ρ=0.2 harmful (+0.046), ρ=0.3 neutral (+0.002), ρ=0.4 beneficial (−0.023)—shows a graceful transition rather than abrupt failure, consistent with the detector's signal-to-noise ratio gradually crossing the threshold needed for reliable α_{eff} estimation.

R Statistical Power Analysis

The adaptive-vs-uniform comparison at ρ=0.4 has Cohen's $d = \bar{\Delta}/\text{SD}_\Delta = 0.023/0.006 = 3.83$ (5 seeds, paired). A one-sided paired t-test at α=0.05 with d=3.83 and N=5 achieves statistical power >0.99 (noncentrality $\lambda = d\sqrt{N} = 8.56$, $t_{0.05,4} = 2.13$), indicating that N=5 is more than adequate to detect an effect of this magnitude.

The low cross-seed variance (σ=0.006) reflects the experimental design: seeds affect only parameter initialization, not data order (the dataloader is deterministic), so the primary stochastic axis is initialization noise at 0.6B scale. The uniform baseline exhibits comparable variance (σ=0.008), confirming that low dispersion is a property of the experimental setting rather than a method artifact.

We apply Bonferroni–Holm correction for k=9 family-wise comparisons (six reweighting methods × one contamination level, plus three additional ρ levels). After correction, p_{BH}=0.011 at ρ=0.4, reaching conventional significance at α=0.05. At ρ=0.6 (3 seeds), p_{BH}=0.064; the family includes structurally heterogeneous tests (different methods, different ρ), which makes Bonferroni–Holm conservative. The uncorrected p-values (0.0012 and 0.010 respectively), directional consistency across all seeds, and the dose-response pattern across ρ levels collectively support a real effect.

S Supplementary Analyses Summary

Analysis	Key result	App.
Downstream tasks	No gain (5 benchmarks, $\lvert\Delta\rvert \leq 0.01$)	I
Cross-model (Llama)	Mixed (MMLU +1.6, HellaSwag −0.6)	O
τ sweep	[0.5, 0.8] all improve; τ>0.8 degrades	D
ρ=0.3 boundary	Δ=+0.002 (neutral, 3 seeds)	Q
Statistical power	Cohen's d=3.83, power >0.99 at N=5	R

Table 17: Summary of supplementary analyses.

T Use of AI Assistants

We used Claude (Anthropic) as an AI coding and writing assistant throughout this work. Specifically:

- **Code:** drafting experiment scripts, data-processing pipelines, and analysis utilities; debugging and refactoring.
- **Writing:** drafting and revising paper sections, polishing prose, and reformatting LaTeX.
- **Literature:** summarizing related papers to inform the related-work section.